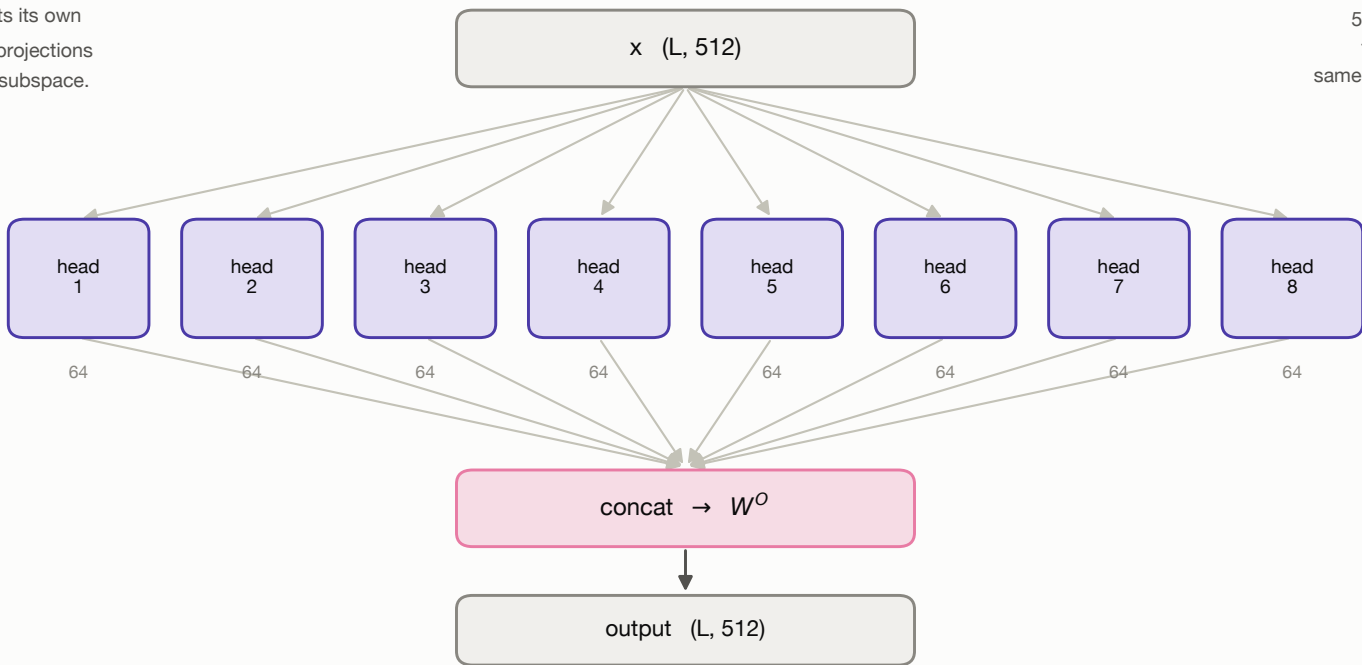


Multi-head attention: h parallel subspaces at one head's cost

Each head gets its own W^Q, W^K, W^V projections into a 64-dim subspace.



$512 = 8 \times 64$, so the split is free: same total compute.

One softmax can attend to essentially one place. Translating "the red house" as "la casa roja" needs the noun tracked for gender agreement and the adjective tracked for reordering — at the same time.